

Bench-testing your logic and calibrating your sensor on a desk *before* soldering permanent high-voltage AC lines is the smartest engineering decision you can make.

For this simplified bench test, we will temporarily strip away all the high-voltage AC protection (fuses, MOVs, snubbers, and 220V relays). We will power only the **ESP32-C3 SuperMini**, the **HLK-LD2410B sensor**, and your **HLK-PM01 5V power supply** (or a simple USB cable), using the Arduino Serial Monitor to visually prove that your presence detection and 3-second timer are working flawlessly.

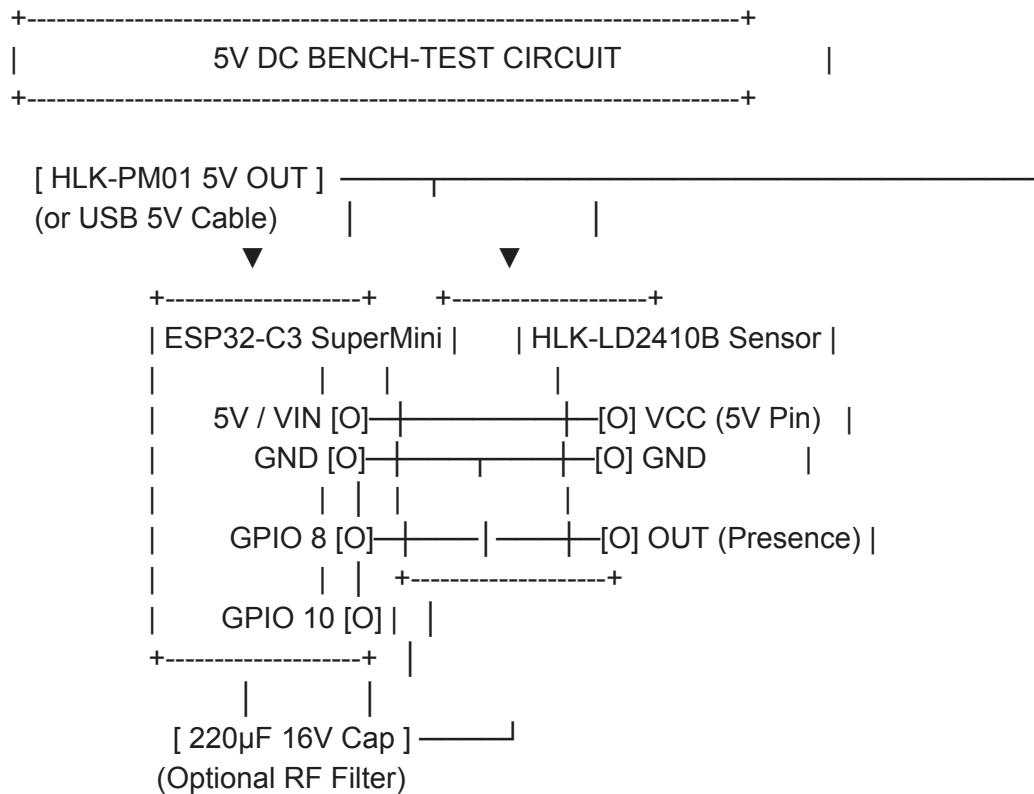
Step 1: Simplified Bench-Test Pinout Table

You only need **three wires** between the sensor and the microcontroller for this test: power (5V), ground (GND), and the digital presence signal (OUT).

Component Pin	ESP32-C3 SuperMini Pin	Connection Purpose
HLK-PM01 +5V (or USB 5V)	5V / VIN	Supplies 5V DC to the ESP32-C3 and LD2410B.
HLK-PM01 GND (or USB GND)	GND	Common system ground rail.
HLK-LD2410B VCC	5V / VIN	Must be 5V. Powers the 24GHz radar oscillator.
HLK-LD2410B GND	GND	Common ground rail.
HLK-LD2410B OUT	GPIO 8	Digital presence signal (HIGH = Human detected, LOW = Empty).
Test Output / LED (Optional)	GPIO 10	This is your simulated "Light Bulb/Relay" pin.

Pro-Tip for Power Sag: If your ESP32-C3 restarts or bootloops when the radar fires, place your **220µF 16V capacitor** directly across the 5V and GND pins on your breadboard to absorb the RF current spike.

Step 2: Simplified Bench-Test Connection Diagram



Step 3: Standalone Bench-Test Firmware (ESP32-C3)

This lightweight, standalone test sketch removes all network complexity (Wi-Fi, MQTT, ESP-NOW) so you can focus 100% on verifying hardware responsiveness and your **3-second off-delay timer**. It prints clean, real-time status updates directly to your Arduino Serial Monitor.

```
// --- PIN CONFIGURATION FOR BENCH TEST ---
#define RADAR_OUT_PIN 8 // Connected to HLK-LD2410B OUT Pin[cite: 1]
#define SIMULATED_LIGHT_PIN 10 // Simulates the Bathroom Relay/Light[cite: 1]

// --- TIMING & STATE VARIABLES ---
bool lightIsActive = false;
unsigned long lastPresenceTimestamp = 0;
const unsigned long OFF_DELAY_MS = 3000; // Exact 3-second off-delay

void setup {
  Serial.begin(115200);
```

```

delay(1500); // Give Serial Monitor time to initialize

Serial.println("=====");
Serial.println(" HLK-LD2410B + ESP32-C3 BENCH CALIBRATION TEST ");
Serial.println("=====");

// Configure Pins
pinMode(RADAR_OUT_PIN, INPUT_PULLDOWN); // Pull-down prevents floating pin noise
pinMode(SIMULATED_LIGHT_PIN, OUTPUT);
digitalWrite(SIMULATED_LIGHT_PIN, LOW); // Start with simulated light OFF

Serial.println("[SYSTEM READY] Waiting for human presence...");
}

void loop {
  // Read digital logic from LD2410B OUT pin
  bool humanPresent = (digitalRead(RADAR_OUT_PIN) == HIGH);

  if (humanPresent) {
    // Continuously refresh the timestamp while a human is detected
    lastPresenceTimestamp = millis();

    if (!lightIsActive) {
      digitalWrite(SIMULATED_LIGHT_PIN, HIGH);
      lightIsActive = true;
      Serial.println("-----");
      Serial.println("[STATUS: HIGH] Human Detected -> LIGHT TURNED ON ");
      Serial.println("-----");
    }
  } else {
    // If no human is present, check if 3 full seconds have elapsed
    if (lightIsActive && (millis() - lastPresenceTimestamp >= OFF_DELAY_MS)) {
      digitalWrite(SIMULATED_LIGHT_PIN, LOW);
      lightIsActive = false;
      Serial.println("-----");
      Serial.println("[STATUS: LOW] Room Empty -> LIGHT TURNED OFF ");
      Serial.println("-----");
    }
  }
}

delay(20); // Small loop stabilization delay
}

```

Step 4: Step-by-Step Bench Calibration & Verification Protocol

1. Upload & Open Serial Monitor

- Plug your ESP32-C3 SuperMini into your computer via USB.
- In the Arduino IDE, select your board and COM port, then **Upload** the sketch.
- Open **Tools > Serial Monitor** and set the baud rate to **115200**.

2. Pair with the HLKRadarTool App

- Power up the sensor and open the **HLKRadarTool** app on your phone (ensure Bluetooth and GPS/Location are enabled).
- Scan and connect to **HLK-LD2410B_XXXX** using the default password **HiLink**[cite: 1, 2].

3. Calibrate Your Distance Gates on Your Desk

While watching the app's live target graph:

1. **Set the Maximum Gate:** Change both **Max Moving Gate** and **Max Static Gate** to **3**. This restricts detection to roughly 2.25 meters, ensuring it won't pick up movement through walls.
2. **Tune Static Sensitivity:** Look at **Gate 2** (the primary still zone). Set **Static Sensitivity** to **15** or **20**. Sit completely still in front of your desk sensor and breathe normally—verify that the app's blue "Static Energy" bar stays above your threshold line.
3. **Set Unmanned Duration:** In the app settings, set **Unmanned Duration** to **5 seconds**.

4. The Final Bench-Test Verification

1. **Wave your hand / Sit in front of the sensor:**
 - Your Serial Monitor should instantly print:
[STATUS: HIGH] Human Detected -> LIGHT TURNED ON
2. **Stand up and walk out of the room (or point the sensor at an empty wall):**
 - Watch your clock. Once the radar's internal unmanned timer clears, the ESP32 will count exactly **3 seconds**[cite: 1, 2], and your Serial Monitor will print:
[STATUS: LOW] Room Empty -> LIGHT TURNED OFF

Once you see this cycle repeating cleanly on your desk without bootloops or false triggers, your sensor is officially calibrated and ready to be transferred to your permanent Zero PCB build!